# KCM Heat-Shock Transformation



**PROTOCOL**

1. **Get your plates warming first — it is slow.** Check they carry the antibiotic(s) on your labsheet, then put them in the **incubator**. Cold plates carry condensation and **you cannot write on a wet plate**.
2. Turn on the **EchoTherm**: **block A 4 °C**, **block B 42 °C**. Wait until both are at temperature and leave them there.
3. Once warm and dry, **label the bottom** of each plate: date, initials, strain, plasmid, selection.
4. Put a **100 µL** cell aliquot on block A, thaw ~30 s, add **25 µL KCM**, pipette gently. One aliquot does **3 reactions**.
5. Put the DNA tube on block A and let it cool **30 s**.
6. Add **40 µL** of the cell/KCM mix to the DNA. Mix gently.
   – **40 µL to 10 µL DNA** assumes DNA is ~20% of the total. Too much DNA dilutes the salts and **drops the efficiency**.
   – Large reaction (~20 µL): the whole tube of cells. Retransforming a miniprep: **0.5 µL** plasmid into 10 µL cells.
7. Hold on block A: **4 °C for 10 min**.
8. Move to block B: **42 °C for 90 s**.
9. Back to block A: **4 °C for 1 min**.
10. **If your selection is anything other than Amp/Carb, rescue first:** add **200 µL 2YT**, move to a 1.5 mL tube, and shake at 37 °C for **1 h**. Amp/Carb plates directly.
11. Plate everything. Incubate **inverted** at 37 °C overnight, and **cancel the temperature programs**.